

Appendix B

Commonly you would find that the application/software package that you wish to install is not available as a deb/rpm or as binary package of your choiced distribution or your platform, in that event your only option is to install from source. Installing from source all has certain advantages, which shall be especially crucial for an advanced Linux user, the primary being the fact that application is optimised for your machine and it performs in more stable and fast manner, since while compiling and linking it takes in the parameter for your specific machine into account to generate the final bytecode. On the downside, installing from source requires you to satisfy a lot more dependencies, most of which you will have to manually install. For example, if you require libXYZ for an application to work, you would also require libXYZ-dev for it compile on your machine.

Before we proceed any further, let us prepare our machine for the task by installing a few common necessities.

```
$ sudo apt-get install build-essential checkinstall
```

followed by

```
$ sudo apt-get install cvs subversion git-core mercurial
```

The first command is for the necessary tools for compiling and linking from source, the second one is for installing the tools that are required for obtaining the source code.

Now the first step towards installing applications from source is to obtain the source code. There are two popular ways to get it, one as a simple compressed archive download from the project web page on the internet and the second is to download it from the version control system. The second method delivers more up-to-date packages and once you get a hang of it, it is also the easier way.

If you have downloaded the package from the internet as a compressed archive, you would need to extract the contents. Typically the packages will be a tar.gz file. We are listing the instructions for tar.gz, tar.bz2 and zip, which should cover 99.5 per cent of all packages found.

For tar.gz

```
$> tar -zxvf <filename>
```

For tar.bz2

```
$> tar -jxvf <filename>
```